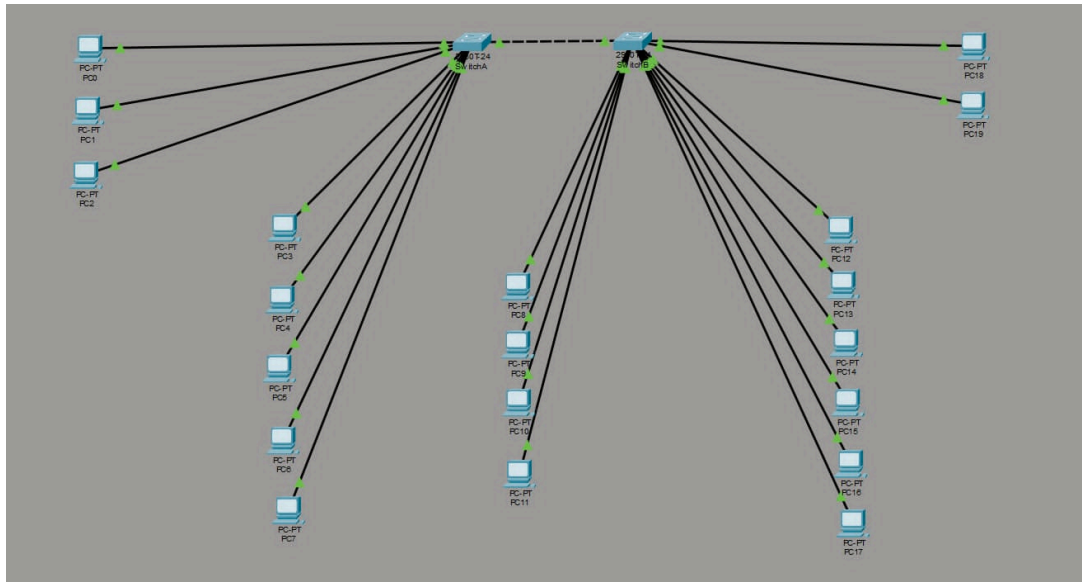


SCREENSHOT W/DISCUSSION



```
SwitchA
Physical Config CLI Attributes
%LINK-S-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-S-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up

Switch>enable
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                           Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                           Gig0/2
10   CCIS                    active    Fa0/2, Fa0/3, Fa0/4
20   INS                     active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9
30   IISM                    active
40   APDEV                   active
50   NETAD                   active
1002 fddi-default           active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active

Switch#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,50

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,50

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,50
```

```
SwitchB
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>
Switch>enable
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                           Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                           Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                           Gig0/2
10   CCIS                    active
20   INS                     active
30   ITSM                    active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
40   APDEV                   active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/11
50   NETAD                   active    Fa0/12, Fa0/13
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active

Switch#show interfaces trunk
Port      Mode          Encapsulation  Status        Native vlan
Fa0/1     desirable     n-802.1q       trunking      1

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,50

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,50

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,50
```

The network topology shown in the figure consists of two Layer 2 switches (Switch A and Switch B) connected through a trunk link. Multiple PCs are connected to each switch and are grouped into different VLANs based on their respective departments. Each VLAN represents a separate logical network, allowing devices within the same department to communicate while remaining isolated from other departments.

VLANs were created and assigned to specific switch ports to organize users logically rather than physically. This setup improves network management, enhances security, and reduces unnecessary broadcast traffic. The trunk link between the two switches enables VLAN information to pass between them, allowing devices in the same VLAN but connected to different switches to communicate successfully.

A Class B IP addressing scheme (172.16.0.0/16) was used, with each VLAN assigned its own subnet. This ensures proper IP organization and avoids address conflicts. Since the network only uses Layer 2 switches and does not include a router or Layer 3 switch, inter-VLAN communication is not enabled. As a result, devices can only ping other devices within the same VLAN, which is expected behavior in a VLAN-segmented network.

Overall, this configuration demonstrates effective VLAN implementation, proper trunking between switches, and logical separation of departments within the network.

SOURCE CODE

Assign

VLAN ID	Department	Network
10	CCIS	172.16.10.2 - 4
20	INS	172.16.20.2 - 6
30	ITSM	172.16.30.2 - 5
40	APDEV	172.16.40.2 - 7
50	NETAD	172.16.50.2 - 3

Ports

Switch A

CCIS	Fa0/2 - 0/4
INS	Fa 0/5 - 0/9

Switch B

ITSM	Fa 02 - 0/5
APDEV	Fa 0/6 - 0/11
NETAD	Fa 0/12 - 0/13

Create VLANs

SWITCH A & B

```
enable
configure terminal
```

```
vlan 10
  name CCIS
exit
```

```
vlan 20
  name INS
exit
```

```
vlan 30
```

```
    name ITSM
exit

vlan 40
    name APDEV
exit

vlan 50
    name NETAD
Exit

end
write memory
```

Assign Access Ports to VLANs

SWITCH A

```
enable
configure terminal

! CCIS - VLAN 10
interface range fa0/2 - 4
    switchport mode access
    switchport access vlan 10
exit

! INS - VLAN 20
interface range fa0/5 - 9
    switchport mode access
    switchport access vlan 20
exit
```

SWITCH B

```
enable
configure terminal

! ITSM - VLAN 30
interface range fa0/2 - 5
    switchport mode access
    switchport access vlan 30
exit
```

```
! APDEV - VLAN 40
interface range fa0/6 - 11
  switchport mode access
  switchport access vlan 40
exit

! NETAD - VLAN 50
interface range fa0/12 - 13
  switchport mode access
  switchport access vlan 50
exit
```

Trunking

On Switch A

```
configure terminal
interface fa0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,40,50
exit
```

On Switch B

```
configure terminal
interface fa0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,40,50
exit
```

Verify

```
show vlan brief
show interfaces trunk
```